

Christian Scaff

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Research Interests

Compilers • Software Systems • Programming Languages • Neuro-symbolic AI

Education

Columbia University <i>M.S. in Computer Science</i>	Sept 2025 – Dec 2026 GPA: 3.56/4.0
University of Florida <i>B.S. in Computer Science</i>	Aug 2021 – May 2025 GPA: 3.96/4.0

Relevant Coursework: System on Chip Platforms, Operating Systems, Database Systems, Programming Language Concepts, Neural Networks and Deep Learning

Experience

Graduate Research Assistant <i>Columbia University - Barnard PL Laboratory</i>	Sept 2025 – Present <i>New York, NY</i>
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- Demonstrated the use of program synthesis to solve graph-based pursuit-evasion games through an implementation in Temporal Stream Logic (TSL).
- Implemented a TextMate grammar for the TSL specification language.

Research Intern <i>Georgia Tech Research Institute</i>	May 2025 – Aug 2025 <i>Atlanta, GA</i>
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- Developed Wy-Graph, a Python tool that translates LLVM IR programs into knowledge graphs, conveying program semantics and syntax through 28 node types and 41 edge types.
- Wy-Graph is aimed at dataset creation for exploring the applications of neuro-symbolic AI, particularly differentiable inductive logic programming, in program analysis.

Undergraduate Research Programmer <i>University of Florida GATAS Laboratory</i>	Aug 2024 – May 2025 <i>Gainesville, FL</i>
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- Developed four domain-specific languages (DSLs) for AlgebraicJulia, an open-source Julia software ecosystem for applied category theory.
- Converted three handwritten DSLs for ACSets.jl, Catlab.jl, and DiagrammaticEquations.jl into parsing expression grammars (PEGs), improving their maintainability and accessibility.
- Implemented a DSL for cellular sheaf creation in AlgebraicOptimization.jl with MLStyle.jl, enabling the definition of cellular sheaf objects in a more intuitive and mathematical manner.

Undergraduate Teaching Assistant <i>University of Florida CISE Department</i>	Jan 2024 – Dec 2024 <i>Gainesville, FL</i>
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- Assisted with COP 4020: Programming Language Concepts, a course on programming language design and implementation.
- Held five hours of office hours per week, assisting students with programming assignments and conceptual understanding.
- Proctored quizzes, graded assignments, and monitored potential academic dishonesty in student repositories.

Technology Architecture Analyst <i>Accenture</i>	Jun 2024 – Aug 2024 <i>Chicago, IL</i>
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- Automated IT monthly reporting for a healthcare client via Power BI, reducing hours of manual analysis.
- Analyzed five months of CMDB data from ServiceNow and consolidated insights into an interactive Power BI dashboard.
- Monitored incident trends, change requests, and SLA metrics across onshore/offshore teams.

Publications

Mining Beyond the Booleans: Learning Data Transformations and Temporal Specifications
Sam Kouteili, Will Fishell, Christian Scaff, Mark Santolucito, and Ruzica Piskac. — Under submission (CAV 2026)

Open Source Contributions

AlgebraicJulia: An Ecosystem of Software Based on Generalized Algebra and Category Theory in Julia.

1. **ACSets.jl:** <https://github.com/AlgebraicJulia/ACSets.jl>
2. **AlgebraicOptimization.jl:** <https://github.com/AlgebraicJulia/AlgebraicOptimization.jl>
3. **Catlab.jl:** <https://github.com/AlgebraicJulia/Catlab.jl>
4. **DiagrammaticEquations.jl:** <https://github.com/AlgebraicJulia/DiagrammaticEquations.jl>

Wyvern Graph: An LLVM IR to Knowledge Graph Translator.

1. <https://github.com/cscaff/Wyvern-Graph>

Technical Skills

Languages: Julia, Python, C/C++, Java, TypeScript, SQL (MariaDB), Portuguese

Tools: LLVM, SystemC, Siemens Catapult HLS, PEG.jl, MLStyle.jl, Pandas, NumPy, Matplotlib, Scikit-learn, React, Git

Leadership

Vice President

Aug 2024 – May 2025

University of Florida Brazilian Portuguese Club

Gainesville, FL

- Collaborated with executive board to oversee weekly Portuguese conversation hours as well as plan cultural events.

Treasurer

Aug 2023 – May 2024

University of Florida Brazilian Portuguese Club

Gainesville, FL

- Managed \$1,000 in club finances, overseeing spending and reporting to the Student Government.